Materials Chemistry and Physics 254 (2020) 123435

Contents lists available at ScienceDirect

Materials Chemistry and Physics

journal homepage: www.elsevier.com/locate/matchemphys

Effect of functional groups of self assembled monolayer molecules on the performance of inverted perovskite solar cell

Emre Arkan [a], Eyup Yalcin [a], Muhittin Unal [b], M. Zeliha Yigit Arkan [a], Mustafa Can [c,**], Cem Tozlu [a,***], Serafettin Demic [a,d,*]

[a] Department of Materials Science and Engineering, Izmir Katip Celebi University, Izmir, Turkey
[b] Department of Energy Systems Engineering, Karamanoglu Mehmetbey University, Karaman, Turkey
[c] Department of Engineering Sciences, Izmir Katip Celebi University, Izmir, Turkey
[d] Central Research Laboratories, Izmir Katip Celebi University, Izmir, Turkey

HIGHLIGHTS

- Five novel SAM molecules having various functional groups are presented to treat ITO surface.
- SAM molecules have chemically attached to ITO surface and revealed permanent changes in the work function of ITO.
- SAM molecules with different terminal groups have created different interactions with perovskite layer.
- Terminal groups with electron donating moieties and halogen atom have indicated more prominent results.
- Terminal groups with electron withdrawing moieties have indicated deterioration in device parameters.

ARTICLE INFO

Keywords:
Perovskite solar cell
Self-assembled monolayer
Electron donating group
Electron withdrawing group

ABSTRACT

Organic-inorganic halide structure such as hybrid perovskite materials has been appeared as a pioneering approach to be used as a light harvester for cost-effective photovoltaic devices. Since light-absorber material is sandwiched between hole and electron transport layers, interfacial engineering starts playing significant role to develop high efficient perovskite solar cell (PSCs). Specifically, poly(3,4-ethylenedioxythiophene)-poly (styrenesulfonate) (PEDOT:PSS) is used commonly as both electrode modifier material and hole transport layer in inverter type device architecture, but it also suffers from instability of PEDOT:PSS due to its ionic nature. Therefore, self-assembled monolayer (SAM) technic is regarded as a proper approach to overcome this problem. In this work, we present five novel SAM molecules with a feasible methodology to compare effect of electron donating and withdrawing terminal groups on the efficiency of inverted PSCs. Depending on the end group, SAM customization indicates a change in the work function of indium tin oxide (ITO) electrode, rectification of device parameters and passivation of the surface trap states. The present study fills a gap in the literature by indicating a comparative treatment route to more clearly understand interfacial issues between electrode-organic layers and perovskite structure for the fabrication of efficient inverted PSCs. This is the first study to undertake a longitudinal evaluation of the influence of both electron-donating and withdrawing terminal groups on the efficiency of inverted type PSCs.

1. Introduction

Recently, strong research interest in the field of renewable energy has been focused on the improvement of power conversion efficiency (PCE) and development of low-cost materials for efficient sustainable energy production. In particular, the progresses in photovoltaic energy in the last decade have made organic-inorganic hybrid solar cells a promising candidate for respective researches owing to their low

* Corresponding author. Department of Materials Science and Engineering, Izmir Katip Celebi University, Izmir, Turkey.
** Corresponding author.
*** Corresponding author.
E-mail address: serafettin.demic@ikcu.edu.tr (S. Demic).

https://doi.org/10.1016/j.matchemphys.2020.123435
Received 19 March 2020; Received in revised form 12 June 2020; Accepted 15 June 2020
Available online 11 July 2020
0254-0584/

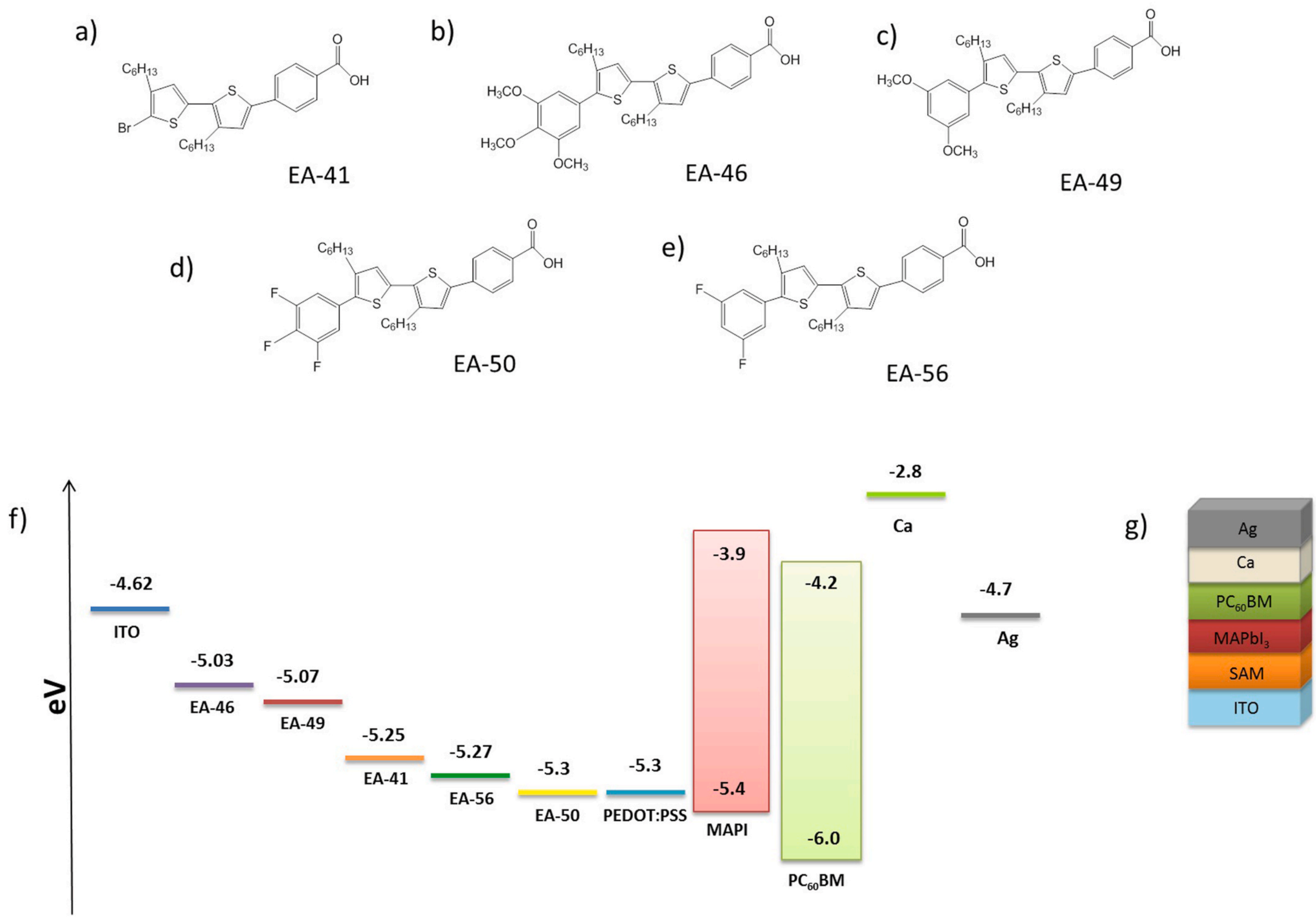

Fig. 1. The molecular structure of SAM molecules (a-e). Energy levels of PSC materials employed in this study (f). Layered structure of SAM based devices (g).

fabrication cost and high PCE [1–5]. A notable example is hybrid perovskites that have emerged as one the most preferred materials for application of optoelectronic systems. On the one hand organometallic halide based perovskite structured materials are known to be vulnerable to moisture, oxygen, UV light, light soaking, heat, electric field, and other possible influences, on the other hand they offer specific features such as strong optical absorption spectrum, long exciton diffusion lengths, high carrier mobilities and sufficient energy band gap [6–8]. In accordance of these attractive material properties, PCE values over 23% have been reported for PSCs [7–12]. ABX_3 is a general chemical formula of organometallic halide based perovskite absorbers in which A is an organic cation (such as CH_3NH_3, NH_2CHNH_2), B is a metallic cation (such as Pb, Sn) and X is halide anion (such as Cl, Br, I). Since methylammonium lead iodide ($MAPbI_3$) demonstrates outstanding electronic and optical characteristics including ambipolar charge properties, large absorption coefficient, and proper band gap of 1.55 eV, it is a broadly preferred constituent of perovskite family [1,13,14].

Device architecture in lead halide PSCs is designed as mesoscopic and planar structures [13–17]. Here, planar devices have significant benefits in terms of low temperature processing and roll to roll fabrication of flexible cells in comparison to mesoscopic structures [17,18]. These devices ensure different charge collection behavior and produced holes and electrons illustrate opposite distributions in *p-i-n* and *n-i-p* planar structures, as well [19]. In comparison with *n-i-p* planar architecture, *p-i-n* planar architecture shows less hysteresis where this particular advantage can be attributed to the cell architecture instead of $MAPbI_3$ itself [20,21].

There is some evidence that interfaces play crucial role in improving solar cell parameters such as open circuit voltage (V_{OC}), short circuit current density (J_{SC}), fill factor (FF) and morphology problems in absorber layer [10,18,22]. Similar to many other organic devices, *p-i-n* type PSCs uses ITO anode contact on account of its high electrical conductivity and optical transparency; however, dangling bonds and surface topography on ITO surface are critical issues for organic devices. Therefore, modifications of ITO surface leading to change in effective work function (WF) and chemical properties of surface play an important role on the solar cell performance [10,23–26]. For example, chemical grafting with organic moieties, UV ozone and oxygen plasma treatments are typical approaches to tailor surface energy and WF [27]. WF tuning via plasma treatments are temporary because it is unstable in atmospheric conditions [28]. Although poly(3,4ethylenedioxylenethiophene):poly(styrenesulfonic acid) (PEDOT:PSS) is a prevalently preferred material used both as a hole transport layer (or electron blocking layer) and for physical modification of ITO surface, its electrical inhomogeneity and acidic nature cause in turn a chemical instability and a decay in electron blocking properties over the time [29–31]. As for chemical grafting, treatment with SAM not only gives rise to chemical modification of ITO electrode but also alters the wettability of substrate surface [26]. Consequently, SAM treatment becomes one of the most appropriate way to improve interface stability of optoelectronic devices [10,32].

It has been illustrated that SAMs can play significant role in regulation interfacial optoelectronic properties and therefore they have been applied to many systems including organic solar cells, dye-sensitized solar cells, organic light emitting diodes, transistors, and etc [33–35]. SAM molecules create ordered molecular alignment inducing permanent

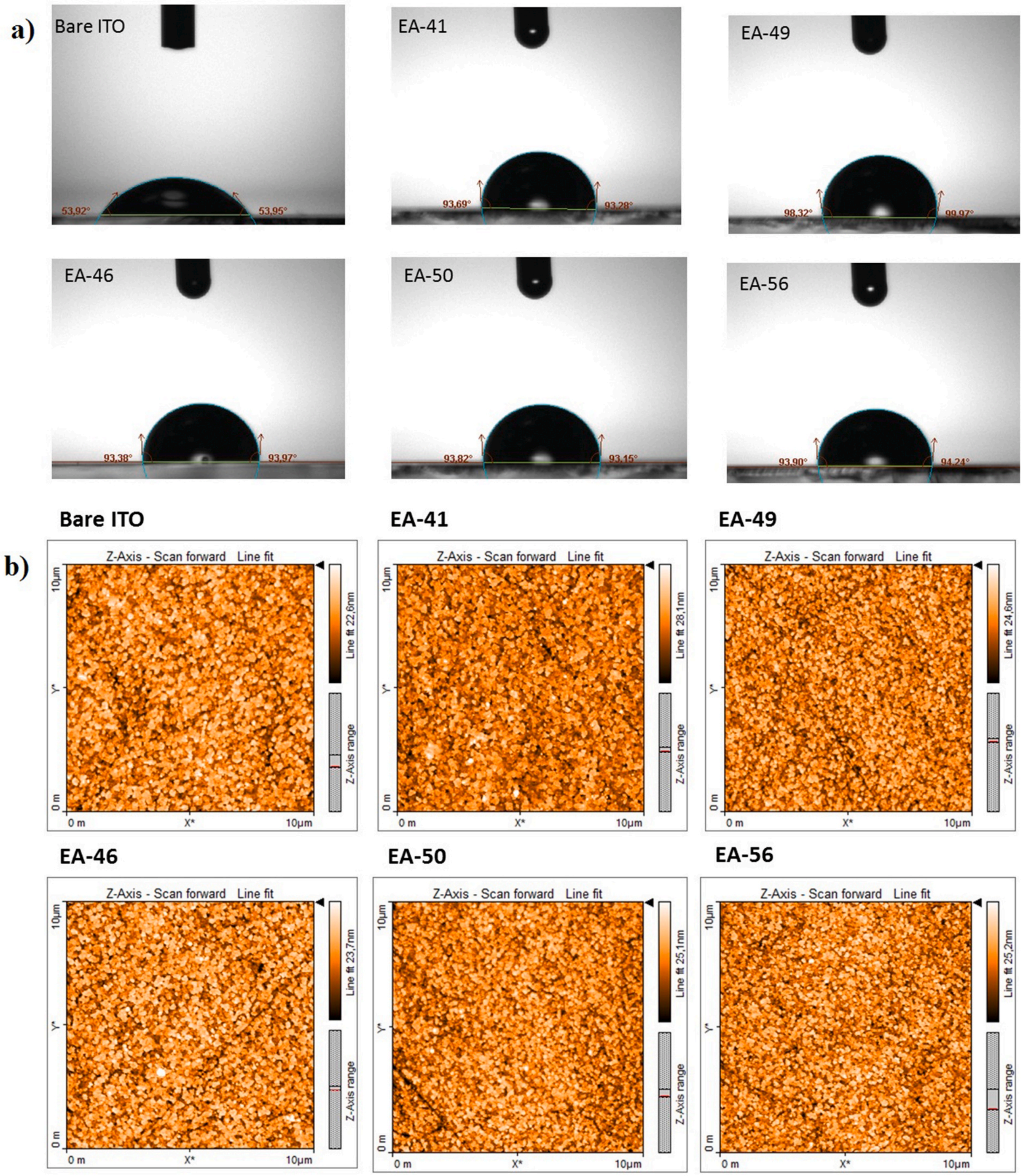

Fig. 2. Changes in water contact angle with regard to bare and different SAM coated ITO substrates (a). Topographical AFM images of bare and different SAM coated ITO substrates (b).

Table 1
Comparison of bare and modified ITO surface properties with different techniques.

Samples	Oxidation Potentials (V)	Work Function (eV)	Contact Angle	RMS (nm)
Bare ITO	–	−4.62	53.93°	3.73
EA-41	1.04	−5.25	93.49°	4.38
EA-46	0.92	−5.03	93.68°	4.45
EA-49	0.96	−5.07	99.15°	3.99
EA-50	1.05	−5.30	93.49°	4.06
EA-56	1.00	−5.27	94.07°	4.27

dipole moment, which is convenient to adjust WF of target substrate [35,36]. Chemical structure of SAM molecules are basically divided into three components so called an anchoring group responsible for chemical attachment on surface, a spacer group used for regulating interfacial features of the molecule, and a terminal group mainly designed to tailor surface properties and interactions at interface with upper layer [37]. In case of their application in PSCs, interfacial properties can be dominated by interfacial chemical interactions because Pb^{+2} ions in perovskite constituent can form several chemical interactions with different type of terminal groups of SAM molecules, which directly affect both the electron density in vicinity part of perovskite film and device parameters such as V_{OC}, FF, and J_{SC} [35,36,38,39]. In addition, their easy bulk

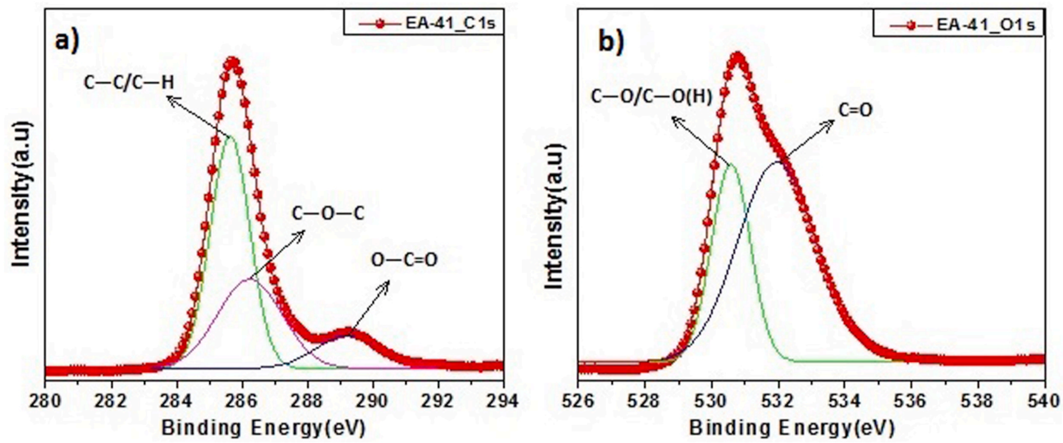

Fig. 3. The XPS high resolution survey spectrum of C1s (a) and O1s (b) for ITO/EA-41.

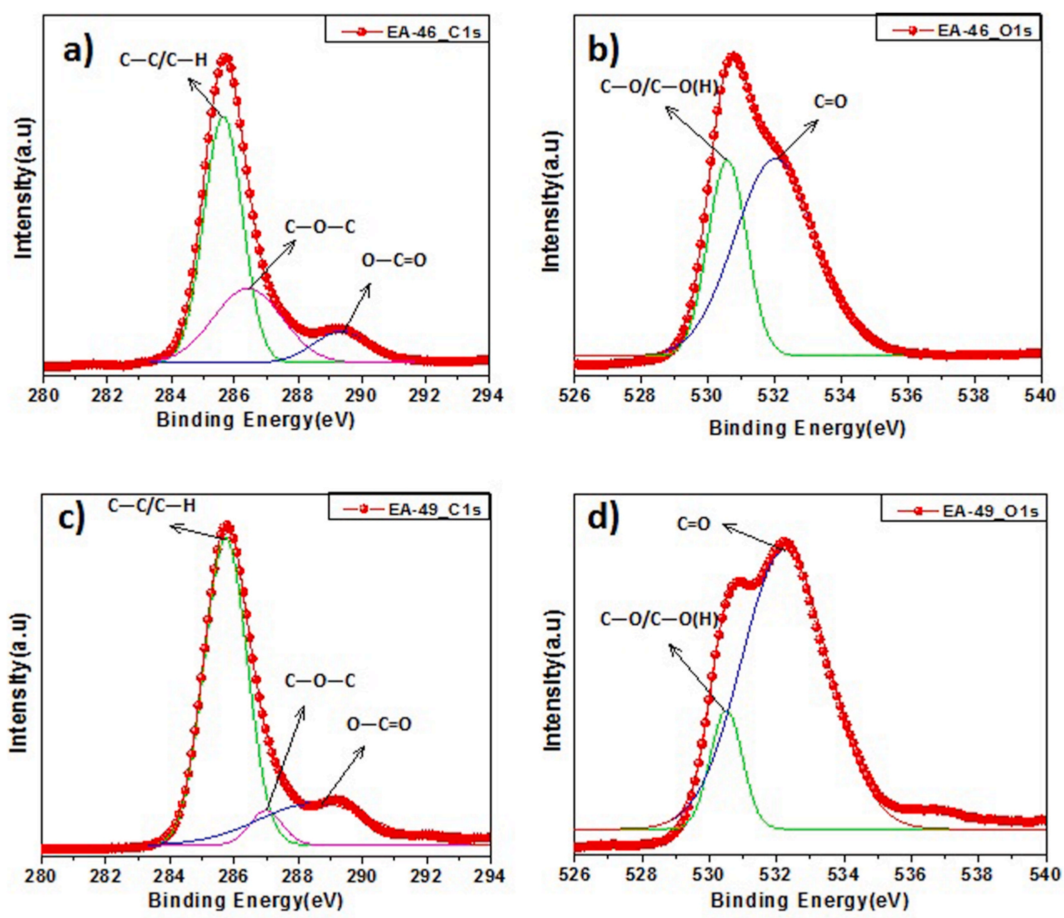

Fig. 4. The XPS high resolution survey spectra of C1s (a, c) and O1s (b, d) for ITO/EA-46 and ITO/EA-49.

synthesis, simple purification and facile deposition make SAM molecules potential materials to be used in organic electronics [40].

In this study, five novel SAM molecules having various electron donating and withdrawing functional groups are presented for the modification of ITO surface. The effect of synthesized SAM molecules on surface characteristics such as wettability and perovskite formation and also on photovoltaic performance of fabricated solar cells was examined. The carboxylic acid is preferred as an attaching group as it is proper building block to form molecular self-assemblies on metallic surface by means of chemisorption [37]. This is the reason that it is particularly useful for the device fabrication in which metal oxides is needed to be utilized. Since proposed SAM alignment is of push-pull type nature, contiguous thiophene rings were used as conjugating units in spacer group due to the fact that thiophene has lower delocalization energy and provide superior conjugation than that of benzene [41]. Furthermore, effect of modification, with such molecular system, on WF, hole-collection and short circuit photocurrent density are discussed. This is the first study to undertake a longitudinal evaluation of the influence of both electron donating and withdrawing terminal groups on efficiency of inverted type PSCs.

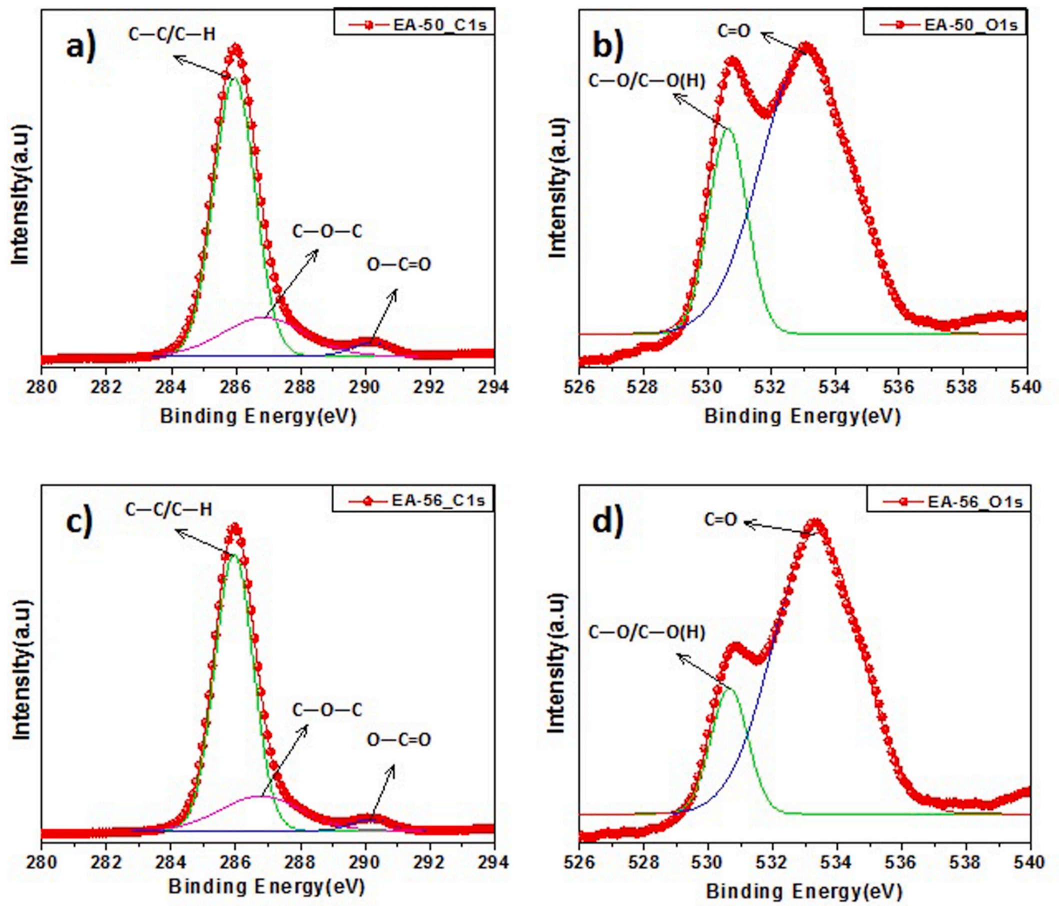

Fig. 5. The XPS high resolution survey spectra of C1s (a, c) and O1s (b, d) for ITO/EA-50 and ITO/EA-56.

Table 2
The functional groups and their related binding energies for ITO/EA-41, ITO/EA-46, ITO/EA-49, ITO/EA-50 and ITO/EA-56.

Samples	O1s (eV)		C1s (eV)			S2p (eV)	F1s (eV)
	C—O/C—O(H)	C=O	C—C/C—H	C—O—C	O—C=O		
ITO/EA-41	530.77	531.96	285.70	286.32	289.24	169.82	–
ITO/EA-46	530.56	532.07	285.66	286.39	289.32	169.58	–
ITO/EA-49	530.52	532.27	285.75	266.90	289.20	165.25	–
ITO/EA-50	530.66	533.08	285.93	286.78	290.17	164.89	689.77
ITO/EA-56	530.64	533.35	285.93	286.78	290.17	164.36	689.31

2. Materials

Methyl 4-bromobenzoate, 3,5-difluorophenylboronic acid, 3,5-dimethoxyphenylboronic acid, [1,1′-bi(diphenylphospino) ferrocene] dichloropalladium(II) ($Pd(dppf)Cl_2$) and poly(3,4-ethylenedioxy thiophene)-poly(styrenesulfonate) (PEDOT:PSS), lead(II) iodide (PbI_2) were purchased from Sigma-Aldrich. (5′-Bromo-3,4′-dihexyl-[2,2′-bithiophen]-5-yl)boronic acid were bought from SunaTech Inc. All solvents such as 1,2-dimethoxy ethane (DME), tetrahydrofuran (THF), ethanol (EtOH), anhydrous *N,N*-Dimethylformamide (DMF), anhydrous isopropyl alcohol, toluene, acetone, dimethyl sulfoxide (DMSO) and hydrochloric acid (HCl), potassium hydroxide (KOH) were purchased from Alfa-Aesar. Potassium carbonate (K_2CO_3) was obtained from Riedel de Haen. Methylammonium iodide (MAI) was obtained from Dysol Inc., $PC_{61}BM$ was purchased from Luminescence Technology Corp. (LUM-TEC). ITO was purchased from KINTEC Company.

Methods: All chemicals, solvents, and reagents were used as received from commercial source without further purification. All glassware was oven-dried and all Suzuki-Coupling reaction was performed under inert (N_2) atmosphere. For further steps such as synthetic procedures, device fabrication process, instrumentation and device characterization sections, Supporting Information can be referred.

3. Result and discussion

In this study, five novel molecules, which have self-assembly forming ability with various functional groups, have been employed for the surface modification of ITO electrode. The chemical structure of molecules and *p-i-n* type device configuration are indicated in Fig. 1. Since the decrease in energy level between WF of ITO electrode and HOMO level of $MAPbI_3$ is essential for increasing the efficient hole collection, the rationale for the modification of ITO surface with SAMs is to closely match the energy barrier alignment of the ITO/$MAPbI_3$ interface.

Various terminal groups with electron donating and withdrawn characteristics are intentionally chosen because, to the best of our knowledge, there is no synergistic approach exists in the literature indicating the effect of a change of terminal group on energy level alignment and performance of perovskite solar cell.

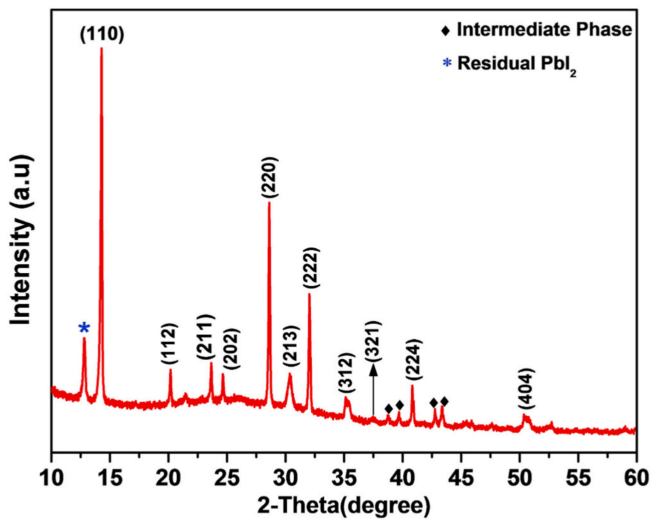

Fig. 6. XRD pattern of $CH_3NH_3PbI_3$ perovskite film.

To assess whether and how surface modification has affected WF of ITO electrode, Kelvin-Probe Force Microscopy (KPFM) measurements before and after surface coverage were implemented. The bare-ITO shows a WF value of 4.62 eV and modified ITO samples exhibit WF values of 5.25, 5.03, 5.07, 5.30, and 5.27 eV for EA-41, EA-46, EA-49, EA-50, and EA56, respectively (Fig. S23). In accordance with energy level alignment of electronic states, an effective photo-carrier collection at the anode side is expected with having WF of ITO electrode closer to HOMO level of $MAPbI_3$ [1,42]. It can be concluded from KPFM analysis that surface modification of ITO by SAMs with electron donating end groups are more favorable in terms of hole collection with respect to modification by SAMs with electron withdrawing end groups. (For conductivity measurements of ITO electrodes, see Fig. S29).

Further information of the key features of SAM molecules with respect to changing terminal groups was gained by examining their electrochemical behavior on ITO surface through cyclic voltammetry. As shown in the voltammograms (Fig. S24), the inflection points of surface modified ITO electrodes have been found to be, in turn, 1.04, 0.92, 0.96, 1.05, and 1.0 for EA-41, EA-46, EA-49, EA-50, and EA-56. It should be noticed in here that in agreement with previous results, current findings demonstrate that existence of methoxy groups on phenyl backbone decreases oxidation potential of SAM molecules and leads to alignment of WF at lower energy level [10,43]. This is a possible explanation for the early inflection point of EA-46 and EA-49 grafted ITO electrodes. This evidence also is in line with CPD (contact potential difference) values gained from KPFM plots and elucidates why EA-46 and EA-49 modified ITO surfaces have lower WF values in comparison to remaining SAMs' modified ITO electrodes.

The contact angle and surface texture are two important parameters, where wetting behavior and topographical properties need to be defined, to evaluate modification of any surface by SAMs. Thus, sessile water droplet experiment was performed to reveal the changes in contact angle values of bare and modified ITO samples. While bare-ITO surface has hydrophilic character, SAMs modified ITO surfaces have higher contact angle values in comparison to bare ITO surface and exhibit hydrophobic character as shown in Fig. 2a. Along with wettability values, surface topography and roughness are synergistically examined because these phenomena are corresponding to wetting properties of sample surface (Fig. 2b). As an overall description, any rough surface becomes more hydrophobic when chemically modified by hydrophobic moieties [44]. As indicated in Table 1, root-mean-square (RMS) values of ITO surfaces are increased after surface modification by SAMs because of the deposition of extra layer [32]. This surface roughness values and their association with contact angle measurements once again bear out that ITO surface are covered by SAMs.

The surface chemical composition of modified ITO electrodes coated with EA-41, EA-46, EA-49, EA-50 and EA-56 molecules were determined by XPS. Fig. 3 shows the high-resolution survey spectra of C1s and O1s to analyze the atomic bonds formed on ITO surface for ITO/EA-41. C1s spectra are fitted by three peaks corresponding to carbon in different chemical environment. The binding energy peaks of C1s for ITO/EA-41 at 285.70, 286.32 and 289.24 eV are assigned to C—C/C—H, C—O—C and O—C=O, respectively. The peak assigned to O—C=O (ester bonding) at 289.24 eV indicates the formation of covalent bonds between —COOH (carboxylic acid) head group of EA-41 (Fig. 3a) molecule and —OH (hydroxyl group) existent on the ITO surface [10,45,46].

O1s high-resolution survey spectrum shows two main components for ITO/EA-41 (Fig. 3b). The O1s of ITO/EA-41 peaks observed at 530.77 and 531.96 eV are associated with lattice oxide in $ln_xSn_yO_z$ (C—O/C—O(H)) and —COOH, respectively [47,48].

The high-resolution survey spectra of C1s and O1s for ITO/EA-46, ITO/EA-49 and ITO/EA-50, ITO/EA-56 are shown in Fig. 4 and Fig. 5, respectively. As in case of ITO/EA-41, identical functional groups and almost the same peaks are obtained for C1s and O1s high-resolution survey spectra of ITO/EA-46, ITO/EA-49, ITO/EA-50 and ITO/EA-56, respectively. Corresponding binding energies of functional groups for ITO/EA-41, ITO/EA-46, ITO/EA-49, ITO/EA-50 and ITO/EA-56 are listed in Table 2. The SAMs are functionalized with electron donating (MeO) and electron withdrawing (F) substituents on phenyl group. The binding energy shift in C1s of XPS graph confirms the existence of these functional groups. The binding energy of C1s in fluorine substituted SAMs (EA-56 and EA-50) was measured as 285.97 eV while it was found as 285.70 eV for EA-46 and EA-49. The shift to greater value in binding energy of C1s is due to decreasing electron density in O—C=O group of EA-46 and EA-49 molecules. The broadening of different peaks is attributed to change in chemical state of SAMs and depends on the substituted elements.

Fig. S25 and Fig. S26 show the wide survey spectra of ITO/EA-41 and ITO/EA-50, respectively. Almost at the same binding energies C, O, F, S, ln and Sn elements are identified on these five ITO/SAM surfaces by XPS. The binding energies in turn at ~487 and ~496 eV indicates the $Sn3d_{5/2}$ and $Sn3d_{3/2}$ while the peaks at ~446 eV and ~454 eV confirm the $ln3d_{5/2}$ and $ln3d_{3/2}$, respectively [49,50]. In addition, the binding energies in turn at ~164, ~285, ~532 and ~690 eV are attributed to S2p, C1s, O1s and F1s elements [51–53].

The XPS wide survey measurement was also performed on bare-ITO to compare SAM-modified ITO substrates (Fig. S27). In this measurement, the absence of N, S and F confirms that ITO substrates were successfully covered by SAM molecules.

As shown in Fig. 6, XRD is used to characterize the structure of $CH_3NH_3PbI_3$ perovskite. The diffraction peaks at 14.22, 20.15, 23.63, 24.64, 28.59, 30.31, 32.03 and 40.82° correspond to (110), (112), (211), (202), (220), (213), (222) and (224) planes of tetragonal phase of $CH_3NH_3PbI_3$ perovskite, respectively [54–58]. The diffraction peak appearing at 12.83° belongs to the residual PbI_2(*). Moreover, diffraction peaks at 38.97, 39.87, 42.95 and 43.50° are related to PbI_2-MAI-DMF intermediate phase (◆) [58].

The quality of the perovskite film has a crucial effect on performance of PSCs. The film quality is determined by grain size, crystallinity, surface coverage etc [59]. Fig. 7 shows the morphology of the perovskite films deposited on bare ITO (a), ITO/PEDOT:PSS (b), ITO/EA-41 (c) and ITO/EA-46 (d), ITO/EA-49 (e), ITO/EA-50 (f) and ITO/EA-56 (g). It is clear from these images that after modification of ITO with PEDOT:PSS and SAM molecules, bigger grain size, pinhole-free, high compact and full surface coverage films (which is desirable for perovskite solar cells) were obtained. (For grain size distributions, see Fig. S30 and Table S1).

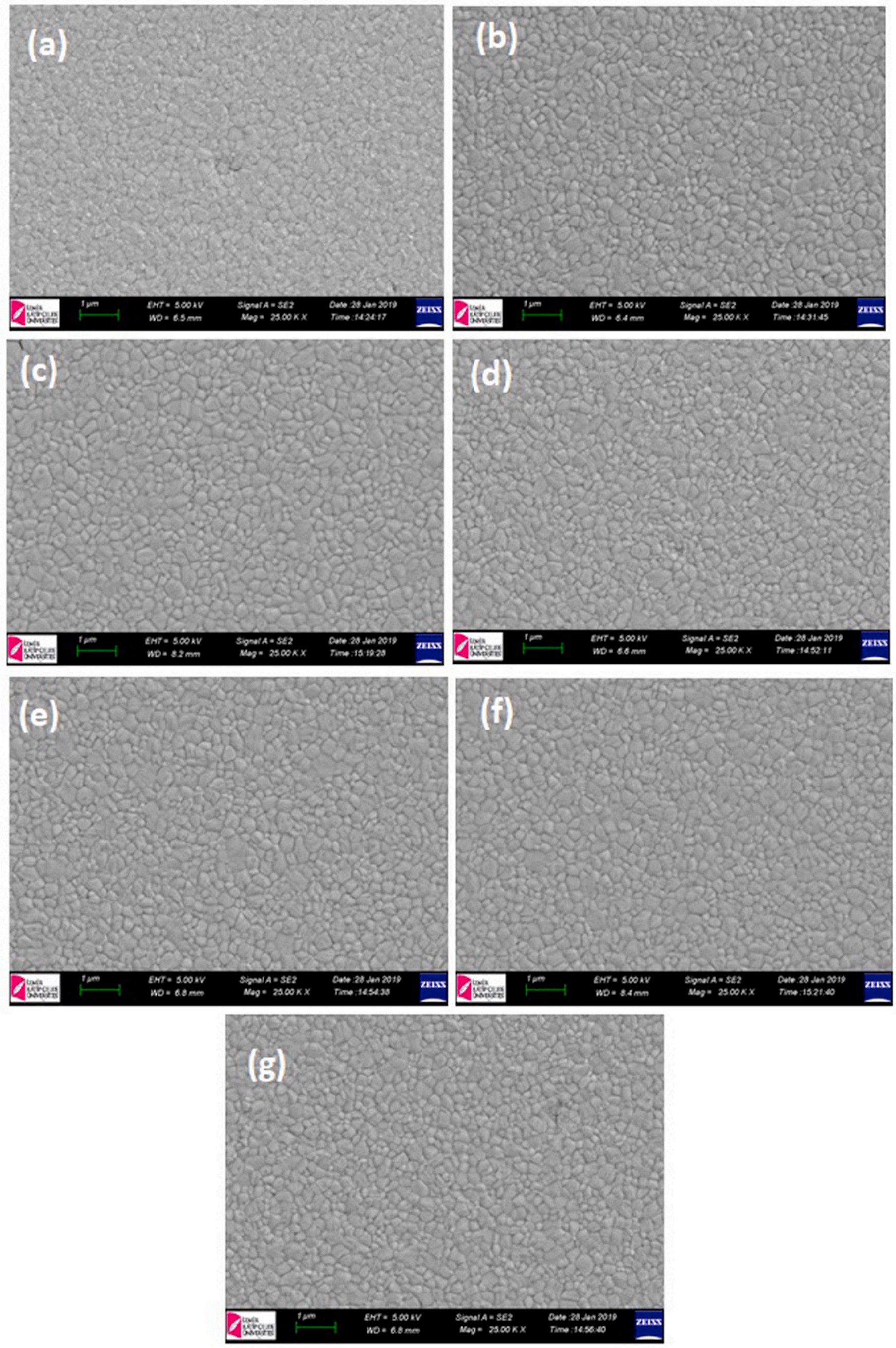

Fig. 7. SEM images of perovskite film deposited on bare ITO (a), ITO/PEDOT:PSS (b), ITO/EA-41 (c), ITO/EA-46 (d), ITO/EA-49 (e), ITO/EA-50 (f) and ITO/EA-56 (g).

3.1. Solar cell performance

The interface modification between ITO and perovskite is essential for the change of energy level alignment that also affects the charge extraction and recombination at interface. This situation can influence device parameter such as the V_{OC}, FF, and J_{SC}.

Fig. 8(a–g) shows current-voltage (I–V) characteristics of solar cells fabricated on bare, SAMs modified and PEDOT:PSS coated ITO substrates and all obtained device parameters are summarized in Table 3, as well. The highest efficiency has obtained from EA-49 modified cell due to the appropriate energy barrier that led to rectification in V_{OC} while its J_{SC} and FF values are comparable with one observed with PEDOT:PSS (the reference cell). Moreover, in agreement with literature, it can be concluded that methoxy terminal group can passivate the trap state in the perovskite layer through coordination bond that can improve optoelectronic properties of perovskite film and therefore resulted in enhancement of the efficiency exhibited by the device [35,36]. However, EA-46 modified ITO electrode has indicated the lower effective WF and thus lower V_{OC} than EA-49 modified PSC because V_{OC} depends on the differences between WF of two electrodes and energy band gap value of perovskite composition in this device arrangement [60–62]. Also, the reduction in FF and J_{SC} can be attributed to its higher serial resistance (Rs) [63]. For this reason, there is a decrease in PCE for EA-46 with respect to EA-49 although their molecular structures are similar.

Additionally, the obtained PCE of EA-41 is more favorable than that of reference cell. As in case of EA-49 treatment, improvement in V_{OC} as

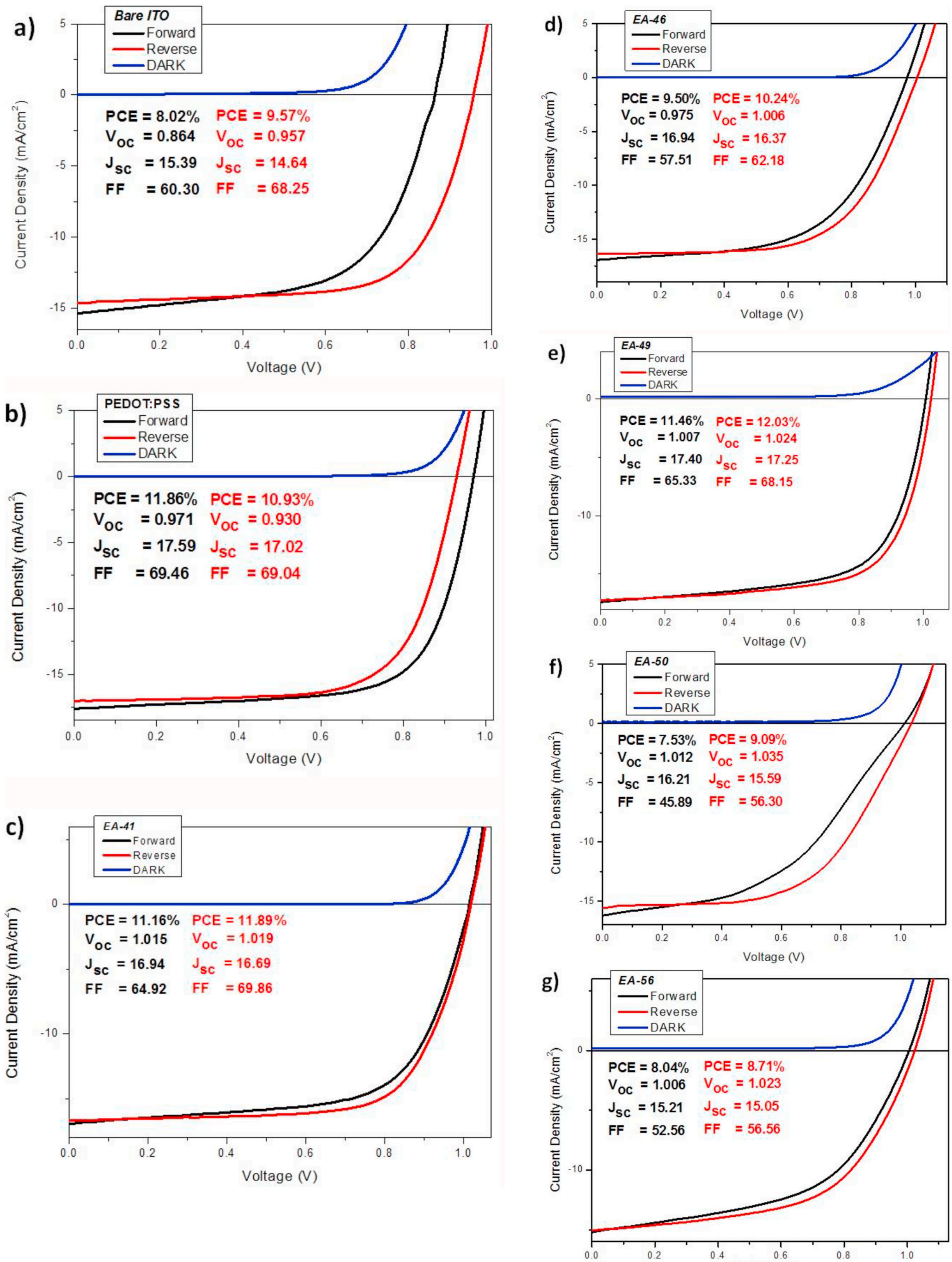

Fig. 8. Current-Voltage characteristics of SAM modified and non-modified devices.

Table 3
Photovoltaic parameters of devices.

Devices	V_{OC} (V)		J_{SC} (mA/cm^2)			FF		PCE (%)		Resistance (Ωcm^2)	
	forward	reverse	forward	reverse	integrated	forward	reverse	forward	reverse	Rs	Rsh
Bare ITO	0.864	0.957	15.39	14.64	15.53	60.30	68.25	8.02	9.57	8.69	335.64
PEDOT:PSS	0.971	0.930	17.59	17.02	16.44	69.46	69.04	11.86	10.93	5.89	681.37
EA-41	1.015	1.019	16.94	16.69	17.33	64.92	69.86	11.16	11.89	9.53	479.32
EA-46	0.975	1.006	16.94	16.37	15.70	57.51	62.18	9.50	10.24	13.15	2271.43
EA-49	1.007	1.024	17.40	17.25	17.95	65.33	68.15	11.46	12.03	5.93	760.04
EA-50	1.012	1.035	16.21	15.59	16.91	45.89	56.30	7.53	9.09	20.95	1242.64
EA-56	1.006	1.023	15.21	15.05	15.70	52.56	56.56	8.04	8.71	13.54	248.91

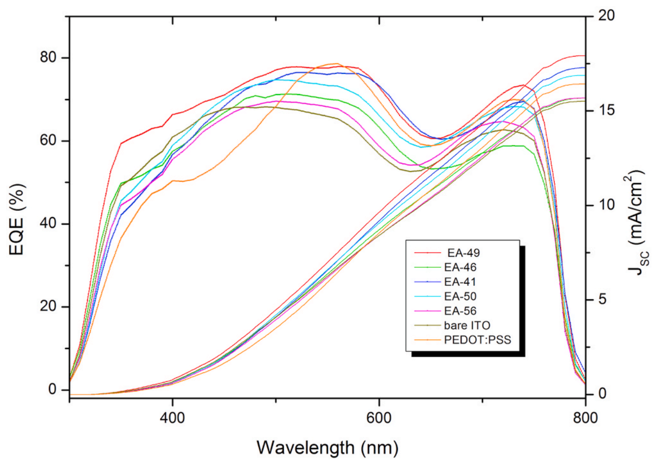

Fig. 9. EQE measurements of devices.

well as intimate interaction between halogen terminal atom and $MAPbI_3$ are closely related to enhance PCE. It should be pointed out that bromine atom can be used to participate to perovskite structure [36,64].

On the other hand, EA-50 and EA-56, bearing acceptor terminal groups, indicate relatively lower PCE compared to other cells. Formation of hydrogen bond between organic constituent of perovskite and fluorine containing molecule has already been introduced to literature [65]. In agreement with this finding, it can be deduced that formation of hydrogen bond (C–H⋯F–C) might withdraw the electron density towards ITO surface and reduce effective charge extraction. Besides, their highest Rs values led to lowest FF and J_{SC} parameters among all ITO-coated devices could be the reason of imperfect contact and resulted in deterioration of PCEs [63]. Their improved V_{OC} values are demonstration of high differences between WFs of their electrodes in device configuration. (For statistical illustration of device parameters, Fig. S31 of Supporting Information can be seen).

It is commonly observed that I–V curve of a perovskite solar cell displays hysteresis. Thus, to understand influence of SAMs' modification on device performance, their effect on hysteresis were investigated. Different from bare-ITO, PEDOT:PSS exhibit less hysteresis whereas all SAMs present negligible or no hysteresis that is in accordance with photovoltaic parameters regardless of scan direction. The reduced hysteresis might be consequence of an efficient hole collection due to decreased interstitial defects of perovskite layer within or near to the interface of materials, which could act as traps for holes and electrons. This interpretation is also in agreement with the theory stating that hysteresis is influenced by the requirement for trap filling and decreased by lower trap density in PSCs [66].

Fig. 9 compares EQE spectra for PSCs with and without ITO modification. As it can be seen, the highest EQE peaks of the best device (EA-49) was achieved to 79.5% photon-to-electron conversion and it has a stronger EQE response than reference cell in the wavelength range of 350–750 nm. This could be consequence of the better charge extraction and recombination reduction owing to optimal chemical interaction at interface [1,35]. The integrated J_{SC} (17.99 mA/cm^2) from the EQE spectra is compliant with that of J_{SC} found (17.40 mA/cm^2) from I–V measurements.

4. Conclusion

In summary, we have synthesized 5 novel SAM molecules to be used in surface modification of ITO electrodes, which then have being utilized to fabricate *p-i-n* type perovskite solar cell. The main structures of proposed molecules have been enriched by flanking thiophene spacer groups to obtain a better conjugation. Since little is known about the influence of different end groups on inverted PSCs, various terminal groups have been employed for the systematic evaluation of their performance in fabricated solar cell. The synthesis of target structures was proved by NMR analysis. KPFM analysis showed that SAM treatment is effective way for the permanent change of the WF of ITO electrode and electron withdrawing terminal groups has led to the higher increase in WF alignment. According to the measurements of CV, contact angle, XPS and AFM images, ITO surface was successfully grafted by SAM treatment. This modification also led to the formation of high quality perovskite films as indicated in XRD analysis and SEM images. Among all suggested SAMs, EA-49 modification exhibited the highest increase in PCE owing to its convenient energy barrier and methoxy terminal group that can passivate the trap states by means of coordination bond. Moreover, EA-41 showed comparable PCE with respect to reference cell due to the intimate interaction between bromine and $MAPbI_3$; therefore, it might be regarded as rival component to PEDOT:PSS. However, SAMs with electron withdrawing terminal groups have drawback of inconvenient energy level alignment and formation of strong hydrogen bond (C–H⋯F–C), which is considered to reduce efficient charge extraction and performance of PSCs. This study offers important insights into various SAM structures and their effect on perovskite to produce feasible alternative to conventional polymers for the fabrication of high performance PSCs.

CRediT authorship contribution statement

Emre Arkan: Investigation, Writing - original draft, Writing - review & editing. **Eyup Yalcin:** Methodology, Software. **Muhittin Unal:** Formal analysis, Data curation. **M. Zeliha Yigit Arkan:** Writing - review & editing, Data curation. **Mustafa Can:** Supervision, Conceptualization. **Cem Tozlu:** Supervision, Writing - review & editing. **Serafettin Demic:** Supervision, Writing - review & editing.

Declaration of competing interest

There are no conflicts to declare.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.matchemphys.2020.123435.

References

[1] D. Akın Kara, K. Kara, G. Oylumluoglu, M.Z. Yigit, M. Can, J.J. Kim, E.K. Burnett, D.L. Gonzalez Arellano, S.m. Buyukcelebi, F. Ozel, Enhanced device efficiency and long-term stability via boronic acid-based self-assembled monolayer modification of indium tin oxide in a planar perovskite solar cell, ACS Appl. Mater. Interfaces 10 (2018) 30000–30007.
[2] J.M. Ball, M.M. Lee, A. Hey, H.J. Snaith, Low-temperature processed meso-superstructured to thin-film perovskite solar cells, Energy Environ. Sci. 6 (2013) 1739–1743.
[3] N.-G. Park, Organometal perovskite light absorbers toward a 20% efficiency low-cost solid-state mesoscopic solar cell, J. Phys. Chem. Lett. 4 (2013) 2423–2429.
[4] P.V. Kamat, Quantum dot solar cells. The next big thing in photovoltaics, J. Phys. Chem. Lett. 4 (2013) 908–918.
[5] M.A. Green, A. Ho-Baillie, H.J. Snaith, The emergence of perovskite solar cells, Nat. Photon. 8 (2014) 506.
[6] G. Han, F. Wang, C. Jia, Y. Chen, M. Guan, H. Liu, Q. Zhang, Enhancement of phase transition temperature from 57° C to 90° C for CH3NH3PbI3 perovskite via SnCl2 doping, Mater. Chem. Phys. 219 (2018) 82–89.
[7] S.K. Ojha, A. Singh, A.K. Ojha, Modifications in structural morphology of CH3NH3PbI3 perovskite using nitrilotriacetic acid and glycine as habit modifiers, Mater. Chem. Phys. 240 (2020) 122149.
[8] A. Kumar, A. Singh, A.K. Ojha, Reshuffling of electronic environment by introducing CH3NH2F+ as an organic cation for enhanced power conversion efficiency and stability of the designed hybrid organic–inorganic perovskite, J. Phys. Chem. C 123 (2019) 13385–13393.
[9] W.S. Yang, B.-W. Park, E.H. Jung, N.J. Jeon, Y.C. Kim, D.U. Lee, S.S. Shin, J. Seo, E. K. Kim, J.H. Noh, Iodide management in formamidinium-lead-halide–based perovskite layers for efficient solar cells, Science 356 (2017) 1376–1379.
[10] E. Yalcin, M. Can, C. Rodriguez-Seco, E. Aktas, R. Pudi, W. Cambarau, S. Demic, E. Palomares, Semiconductor self-assembled monolayers as selective contacts for efficient PiN perovskite solar cells, Energy Environ. Sci. 12 (2019) 230–237.
[11] W.S. Yang, J.H. Noh, N.J. Jeon, Y.C. Kim, S. Ryu, J. Seo, S.I. Seok, High-performance photovoltaic perovskite layers fabricated through intramolecular exchange, Science 348 (2015) 1234–1237.
[12] M.M. Elsenety, M. Antoniadou, A. Kaltzoglou, A.G. Kontos, A.I. Philippopoulos, C. A. Mitsopoulou, P. Falaras, Synthesis, characterization of ((CH3) 3S) 2SnI6-nCln and ((CH3) 3S) 2SnI6-nBrn (n= 1, 2) perovskites and use in dye-sensitized solar cells, Mater. Chem. Phys. 239 (2020) 122310.
[13] J.Y. Jeng, Y.F. Chiang, M.H. Lee, S.R. Peng, T.F. Guo, P. Chen, T.C. Wen, CH3NH3PbI3 perovskite/fullerene planar-heterojunction hybrid solar cells, Adv. Mater. 25 (2013) 3727–3732.
[14] G.E. Eperon, V.M. Burlakov, P. Docampo, A. Goriely, H.J. Snaith, Morphological control for high performance, solution-processed planar heterojunction perovskite solar cells, Adv. Funct. Mater. 24 (2014) 151–157.
[15] H.-S. Kim, C.-R. Lee, J.-H. Im, K.-B. Lee, T. Moehl, A. Marchioro, S.-J. Moon, R. Humphry-Baker, J.-H. Yum, J.E. Moser, Lead iodide perovskite sensitized all-solid-state submicron thin film mesoscopic solar cell with efficiency exceeding 9%, Sci. Rep. 2 (2012) 591.
[16] M.M. Lee, J. Teuscher, T. Miyasaka, T.N. Murakami, H.J. Snaith, Efficient hybrid solar cells based on meso-superstructured organometal halide perovskites, Science 338 (2012) 643–647.
[17] O. Malinkiewicz, A. Yella, Y.H. Lee, G.M. Espallargas, M. Graetzel, M. K. Nazeeruddin, H.J. Bolink, Perovskite solar cells employing organic charge-transport layers, Nat. Photon. 8 (2014) 128.
[18] P. Docampo, J.M. Ball, M. Darwich, G.E. Eperon, H.J. Snaith, Efficient organometal trihalide perovskite planar-heterojunction solar cells on flexible polymer substrates, Nat. Commun. 4 (2013) 2761.
[19] J.H. Noh, S.I. Seok, Steps toward efficient inorganic–organic hybrid perovskite solar cells, MRS Bull. 40 (2015) 648–653.
[20] N.J. Jeon, J.H. Noh, Y.C. Kim, W.S. Yang, S. Ryu, S.I. Seok, Solvent engineering for high-performance inorganic–organic hybrid perovskite solar cells, Nat. Mater. 13 (2014) 897.
[21] J. Seo, S. Park, Y.C. Kim, N.J. Jeon, J.H. Noh, S.C. Yoon, S.I. Seok, Benefits of very thin PCBM and LiF layers for solution-processed p–i–n perovskite solar cells, Energy Environ. Sci. 7 (2014) 2642–2646.
[22] H. Ma, H.L. Yip, F. Huang, A.K.Y. Jen, Interface engineering for organic electronics, Adv. Funct. Mater. 20 (2010) 1371–1388.
[23] W. Feng, S. Rangan, Y. Cao, E. Galoppini, R. Bartynski, E. Garfunkel, Energy level alignment of polythiophene/ZnO hybrid solar cells, J. Mater. Chem. 2 (2014) 7034–7044.
[24] M.C. Scharber, D. Mühlbacher, M. Koppe, P. Denk, C. Waldauf, A.J. Heeger, C. J. Brabec, Design rules for donors in bulk-heterojunction solar cells—towards 10% energy-conversion efficiency, Adv. Mater. 18 (2006) 789–794.
[25] H.L. Yip, S.K. Hau, N.S. Baek, H. Ma, A.K.Y. Jen, Polymer solar cells that use self-assembled-monolayer-modified ZnO/metals as cathodes, Adv. Mater. 20 (2008) 2376–2382.
[26] S. Khodabakhsh, B.M. Sanderson, J. Nelson, T.S. Jones, Using self-assembling dipole molecules to improve charge collection in molecular solar cells, Adv. Funct. Mater. 16 (2006) 95–100.
[27] Y. Zhou, J.W. Shim, C. Fuentes-Hernandez, A. Sharma, K.A. Knauer, A.J. Giordano, S.R. Marder, B. Kippelen, Direct correlation between work function of indium-tin-oxide electrodes and solar cell performance influenced by ultraviolet irradiation and air exposure, Phys. Chem. Chem. Phys. 14 (2012) 12014–12021.
[28] K.H. Lee, H.W. Jang, K.-B. Kim, Y.-H. Tak, J.-L. Lee, Mechanism for the increase of indium-tin-oxide work function by O 2 inductively coupled plasma treatment, J. Appl. Phys. 95 (2004) 586–590.
[29] L.S. Sapochak, A.B. Padmaperuma, X. Cai, J.L. Male, P.E. Burrows, Inductive effects of diphenylphosphoryl moieties on carbazole host materials: design rules for blue electrophosphorescent organic light-emitting devices, J. Phys. Chem. C 112 (2008) 7989–7996.
[30] H. Yan, P. Lee, N.R. Armstrong, A. Graham, G.A. Evmenenko, P. Dutta, T.J. Marks, High-performance hole-transport layers for polymer light-emitting diodes. Implementation of organosiloxane cross-linking chemistry in polymeric electroluminescent devices, J. Am. Chem. Soc. 127 (2005) 3172–3183.
[31] K. Wong, H. Yip, Y. Luo, K. Wong, W. Lau, K. Low, H. Chow, Z. Gao, W. Yeung, C. Chang, Blocking reactions between indium-tin oxide and poly (3, 4-ethylene dioxythiophene): poly (styrene sulphonate) with a self-assembly monolayer, Appl. Phys. Lett. 80 (2002) 2788–2790.
[32] M. Can, A.K. Havare, H. Aydın, N. Yagmurcukardes, S. Demic, S. Icli, S. Okur, Electrical properties of SAM-modified ITO surface using aromatic small molecules with double bond carboxylic acid groups for OLED applications, Appl. Surf. Sci. 314 (2014) 1082–1086.
[33] L. Zuo, Z. Gu, T. Ye, W. Fu, G. Wu, H. Li, H. Chen, Enhanced photovoltaic performance of CH3NH3PbI3 perovskite solar cells through interfacial engineering using self-assembling monolayer, J. Am. Chem. Soc. 137 (2015) 2674–2679.
[34] L. Liu, A. Mei, T. Liu, P. Jiang, Y. Sheng, L. Zhang, H. Han, Fully printable mesoscopic perovskite solar cells with organic silane self-assembled monolayer, J. Am. Chem. Soc. 137 (2015) 1790–1793.
[35] L. Zuo, Q. Chen, N. De Marco, Y.-T. Hsieh, H. Chen, P. Sun, S.-Y. Chang, H. Zhao, S. Dong, Y. Yang, Tailoring the interfacial chemical interaction for high-efficiency perovskite solar cells, Nano Lett. 17 (2016) 269–275.
[36] R. Qiao, L. Zuo, Self-assembly monolayers boosting organic–inorganic halide perovskite solar cell performance, J. Mater. Res. 33 (2018) 387–400.
[37] S.A. Jadhav, Self-assembled monolayers (SAMs) of carboxylic acids: an overview, Cent. Eur. J. Chem. 9 (2011) 369–378.
[38] B. Cao, X. He, C.R. Fetterly, B.C. Olsen, E.J. Luber, J.M. Buriak, Role of interfacial layers in organic solar cells: energy level pinning versus phase segregation, ACS Appl. Mater. Interfaces 8 (2016) 18238–18248.
[39] Q. Wang, C.C. Chueh, T. Zhao, J. Cheng, M. Eslamian, W.C. Choy, A.K.Y. Jen, Effects of self-assembled monolayer modification of nickel oxide nanoparticles layer on the performance and application of inverted perovskite solar cells, ChemSusChem 10 (2017) 3794–3803.
[40] F. Rissner, G.M. Rangger, O.T. Hofmann, A.M. Track, G. Heimel, E. Zojer, Understanding the electronic structure of metal/SAM/Organic− semiconductor heterojunctions, ACS Nano 3 (2009) 3513–3520.
[41] A.K. Jen, V.P. Rao, K. Wong, K.J. Drost, Functionalized thiophenes: second-order nonlinear optical materials, J. Chem. Soc., Chem. Commun. (1993) 90–92.
[42] H. Zhou, Q. Chen, G. Li, S. Luo, T.-b. Song, H.-S. Duan, Z. Hong, J. You, Y. Liu, Y. Yang, Interface engineering of highly efficient perovskite solar cells, Science 345 (2014) 542–546.
[43] S. Amthor, B. Noller, C. Lambert, UV/Vis/NIR spectral properties of triarylamines and their corresponding radical cations, Chem. Phys. 316 (2005) 141–152.
[44] R.N. Wenzel, Resistance of solid surfaces to wetting by water, Ind. Eng. Chem. 28 (1936) 988–994.
[45] P. Prieto, V. Nistor, K. Nouneh, M. Oyama, M. Abd-Lefdil, R. Díaz, XPS study of silver, nickel and bimetallic silver–nickel nanoparticles prepared by seed-mediated growth, Appl. Surf. Sci. 258 (2012) 8807–8813.
[46] K.J. Carroll, J.U. Reveles, M.D. Shultz, S.N. Khanna, E.E. Carpenter, Preparation of elemental Cu and Ni nanoparticles by the polyol method: an experimental and theoretical approach, J. Phys. Chem. C 115 (2011) 2656–2664.
[47] E. Yalcin, D.A. Kara, C. Karakaya, M.Z. Yigit, A.K. Havare, M. Can, C. Tozlu, S. Demic, M. Kus, A. Aboulouard, Functionalized organic semiconductor molecules to enhance charge carrier injection in electroluminescent cell, Opt. Mater. 69 (2017) 283–290.
[48] F. Montagne, J. Polesel-Maris, R. Pugin, H. Heinzelmann, Poly (N-isopropylacrylamide) thin films densely grafted onto gold surface: preparation, characterization, and dynamic AFM study of temperature-induced chain conformational changes, Langmuir 25 (2008) 983–991.
[49] M.S. Raju, S. Bhattacharya, Structural and optical properties of nanocrystalline pure and indium doped tin oxide powders synthesized in a single step by flame spray pyrolysis, Mater. Res. Express 4 (2017), 075034.
[50] Y. Liu, A. Palmieri, J. He, Y. Meng, N. Beauregard, S.L. Suib, W.E. Mustain, Highly conductive In-SnO 2/RGO nano-heterostructures with improved lithium-ion battery performance, Sci. Rep. 6 (2016) 25860.
[51] Z. Ahmad, M.A. Najeeb, R. Shakoor, A. Alashraf, S.A. Al-Muhtaseb, A. Soliman, M. Nazeeruddin, Instability in CH 3 NH 3 PbI 3 perovskite solar cells due to elemental migration and chemical composition changes, Sci. Rep. 7 (2017) 15406.
[52] G. Tai, T. Zeng, J. Yu, J. Zhou, Y. You, X. Wang, H. Wu, X. Sun, T. Hu, W. Guo, Fast and large-area growth of uniform MoS 2 monolayers on molybdenum foils, Nanoscale 8 (2016) 2234–2241.
[53] R. Xu, L. Xiao, L. Luo, Q. Yuan, D. Qin, G. Hu, W. Gan, Nitrogen, sulfur dual-doped mesoporous carbon modified glassy carbon electrode for simultaneous determination of hydroquinone and catechol, J. Electrochem. Soc. 163 (2016) B617–B623.
[54] J. Yu, X. Chen, Y. Wang, H. Zhou, M. Xue, Y. Xu, Z. Li, C. Ye, J. Zhang, P.A. Van Aken, A high-performance self-powered broadband photodetector based on a CH 3 NH 3 PbI 3 perovskite/ZnO nanorod array heterostructure, J. Mater. Chem. C 4 (2016) 7302–7308.

[55] X. Zou, H. Fan, Y. Tian, S. Yan, Synthesis of Cu 2 O/ZnO hetero-nanorod arrays with enhanced visible light-driven photocatalytic activity, CrystEngComm 16 (2014) 1149–1156.
[56] X. Fan, C. Cui, G. Fang, J. Wang, S. Li, F. Cheng, H. Long, Y. Li, Efficient polymer solar cells based on poly (3-hexylthiophene): indene-C70 bisadduct with a MoO3 buffer layer, Adv. Funct. Mater. 22 (2012) 585–590.
[57] J.H. Heo, S.H. Im, J.H. Noh, T.N. Mandal, C.-S. Lim, J.A. Chang, Y.H. Lee, H.-j. Kim, A. Sarkar, M.K. Nazeeruddin, Efficient inorganic–organic hybrid heterojunction solar cells containing perovskite compound and polymeric hole conductors, Nat. Photon. 7 (2013) 486.
[58] X. Guo, C. McCleese, C. Kolodziej, A.C. Samia, Y. Zhao, C. Burda, Identification and characterization of the intermediate phase in hybrid organic–inorganic MAPbI 3 perovskite, Dalton Trans. 45 (2016) 3806–3813.
[59] D. Yang, R. Yang, K. Wang, C. Wu, X. Zhu, J. Feng, X. Ren, G. Fang, S. Priya, S. F. Liu, High efficiency planar-type perovskite solar cells with negligible hysteresis using EDTA-complexed SnO 2, Nat. Commun. 9 (2018) 3239.
[60] P. Swapna, Y.S. Rao, Simulation studies on semiconducting single walled carbon nanotube based bilayer and bulk hetero junction organic solar cells, Mater. Today: Proceedings 2 (2015) 1634–1641.
[61] H. Hoppe, N.S. Sariciftci, Organic solar cells: an overview, J. Mater. Res. 19 (2004) 1924–1945.
[62] S. Günes, H. Neugebauer, N.S. Sariciftci, Conjugated polymer-based organic solar cells, Chem. Rev. 107 (2007) 1324–1338.
[63] C.H. Ng, T.S. Ripolles, K. Hamada, S.H. Teo, H.N. Lim, J. Bisquert, S. Hayase, Tunable open circuit voltage by engineering inorganic cesium lead bromide/iodide perovskite solar cells, Sci. Rep. 8 (2018) 2482.
[64] H.B. Kim, I. Im, Y. Yoon, S. Do Sung, E. Kim, J. Kim, W.I. Lee, Enhancement of photovoltaic properties of CH 3 NH 3 PbBr 3 heterojunction solar cells by modifying mesoporous TiO 2 surfaces with carboxyl groups, J. Mater. Chem. 3 (2015) 9264–9270.
[65] G. Jia, Z.-J. Shi, Y.-D. Xia, Q. Wei, Y.-H. Chen, G.-C. Xing, W. Huang, Super air stable quasi-2D organic-inorganic hybrid perovskites for visible light-emitting diodes, Optic Express 26 (2018) A66–A74.
[66] H.J. Snaith, A. Abate, J.M. Ball, G.E. Eperon, T. Leijtens, N.K. Noel, S.D. Stranks, J. T.-W. Wang, K. Wojciechowski, W. Zhang, Anomalous hysteresis in perovskite solar cells, J. Phys. Chem. Lett. 5 (2014) 1511–1515.